MILITARY SPECIFICATION

CAPSTANS, ELECTRIC (FOR SHIPBOARD USE)

1. SCOPE

1.1 This specification covers electric capstans suitable for installation on Naval ships.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-601 - Boxes, Wood, Cleated Plywood.

MILITARY

JAN-P-72 - Primer, Synthetic, Lacquer-Resisting.
JAN-P-104 - Packaging and Packing for Overseas Shipment - Crates, Sheathed, Wood, Nailed.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood Nailed.
MIL-P-116 - Preservation, Methods of.
JAN-P-125 - Packaging and Packing for Overseas Shipment - Barrier Material, Waterproof, Flexible.
JAN-P-132 - Packaging and Packing for Overseas Shipment - Crates; Unsheathed, Wood, Nailed (for Maximum Net Load of 2,500 Pounds).
MIL-B-233 - Boxes, Repair Parts.
JAN-P-735 - Primer, Paint, Zinc-Chromate, Alkyd Type.
MIL-B-857 - Bolts, Nuts, Studs and Tap Rivets.
MIL-D-963 - Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).
MIL-C-2174 - Controllers and Master Switches, Direct-Current (Naval Shipboard Use).
MIL-C-2212 - Controllers and Master Switches, Alternating-Current (Naval Shipboard Use).
MIL-P-3184 - Packaging of Capstans, Winches, and Windlasses (Shipboard Use), Overseas Shipment.
MIL-F-3541 - Fittings, Lubrication (Hydraulic).
MIL-C-3769 - Crates, Intermediate, Sheathed, Wood, Nailed (for Maximum Net Loads of 3000 Pounds).
MIL-C-3774 - Crates Open, Wood (2,500 to 10,000 Pounds).
MIL-L-10547 - Liners, Case, Waterproof.
MIL-I-15024 - Identification Plates, Information Plates and Marking Information for Identification of Electric, Electronic, and Mechanical Equipment.
MIL-B-15071 - Books, Instruction; Preparation, Contents and Approval.
MIL-S-15083 - Steel, Castings (for Shipboard Applications).
MIL-P-15130 - Paint, Outside, Haze Gray, No. 27 (Formula No. 5H) (for Naval Shipboard Use).
MIL-R-15137 - Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).

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MIL-C-15167 - Compound, Rust-Preventive, Thick-Film (Pigmented and Non-Pigmented).
MIL-S-16113 - Steel, Plate, Structural, Black (Uncoated) and Zinc-Coated (Galvanized).
MIL-P-16298 - Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts (Includes Associated Repair Parts).
MIL-B-16392 - Brake, Electric (Naval Shipboard Use).
MIL-B-16443 - Bronze, Manganese; Castings.
MIL-B-16540 - Bronze, Phosphor; Castings.
MIL-M-17059 - Motors, Alternating Current, Fractional HP (Shipboard Use).
MIL-F-17292 - Format for List of Repair (Spare) Parts for Shipboard Mechanical and Electrical Equipment.
MIL-M-17556 - Motor, Direct-Current, Fractional HP (Shipboard Use).
MIL-P-17652 - Preparation for Delivery of Electronic, Electrical and Electro-Mechanical Parts.
MIL-P-20085 - Packing, Sheet, Plant or Animal Fibre.
MIL-S-20140 - Steel: Forgings for Welding.

NAVY DEPARTMENT

General Specifications for Inspection of Material.
42B5 - Bearings, Ball.
42B10 - Bearings, Roller.

STANDARDS

MILITARY

JAN-STD-19 - Welding Symbols.
MIL-STD-129 - Marking of Shipments.

DRAWINGS

BUREAU OF SHIPS

S2601-860303 - General Type Plan, Capstan and Gypsy Heads, Tabulation of Sizes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply:

AMERICAN GEAR MANUFACTURER'S ASSOCIATION STANDARDS

(Information as to the availability of these standards may be obtained from the American Gear Manufacturer's Association, 302 Empire Building, Pittsburgh 22, Pennsylvania.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

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3. REQUIREMENTS

3.1 Material.- Material shall be entirely suitable for the purpose intended and shall conform to the following specifications as applicable:

3.1.1 Forged steel shall conform to Specification MIL-S-20140.

3.1.2 Cast steel shall conform to Specification MIL-S-15083.

3.1.3 Steel plate shall conform to Specification MIL-S-16113.

3.1.4 Cast iron shall not be used for any part.

3.2 Design.-

3.2.1 Capstans shall be of the reversible type and shall be operable and develop specified line pulls and speeds in both directions of rotation. Each capstan shall consist of a vertical capstan head, shaft, gear reduction, electric motor, controller, electric brake, and all operating accessories necessary for operation independent of anything except power leads. The capstan head, shafting, bearings, gears and bedplate shall be designed to withstand a static load equal to the minimum breaking strength of the specified size rope applied as a bitt to the capstan head at its midheight and normal to its axis. All parts of the capstan shall be designed to withstand a torque causing slippage of the brake or stalling of the driving machinery, whichever is greater. This torque and the static load shall not be considered to act simultaneously. The capstan arrangement shall be one of the following as specified (see 6.1):

(a) Capstan head mounted on weather deck; motor and gear reduction mounted on deck below.
(b) Capstan head mounted on weather deck; motor and gear reduction mounted in underhung position.
(c) Capstan head, gear reduction and motor all mounted on weather deck.

The arrangement shall provide for clear leads at all angles to the capstan head. Pawls will not be required unless specified in the contract or order.

3.2.2 Welded fabrication properly made and stress relieved shall be generally acceptable in lieu of castings. Full description of welding procedure shall be given using welding symbols in conformance with Standard JAN-STD-19.

3.2.3 Design of the capstan shall be such as to eliminate hazards of exposed parts becoming inoperative due to frequent soaking by salt spray and extreme atmospheric conditions occurring on the weather deck of the vessel. Where capstan shafts pass through a deck provision shall be made to obtain thorough watertightness. Flexible couplings or slip joints shall be provided as necessary to compensate for deck deflections.

3.2.4 The unit shall be as compact and light as is consistent with adequate strength and rigidity. Full use shall be made of welded construction to obtain lightest weight.

3.2.5 Positive means shall be provided for securing keys, feathers, splines, nuts, screws, and collars to prevent their working loose. The use of set screws shall be avoided. When set screws are used means shall be provided to prevent their working out.

3.2.6 Capstan head.- The capstan head shall be proportioned as shown on Drawing S2601-860303 in accordance with the rope size specified (see 6.1). The wearing surface shall be hardened to a minimum of 350 Brinell to prevent scoring with use of wire rope.

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3.2.7 Gearing.- All gearing shall be in accordance with AGMA standards, shall have machine cut teeth, and shall be provided with rugged and ample guards for the protection of personnel and the gears. Gears shall be totally enclosed in an oil tight case and shall be lubricated by an oil bath.

3.2.8 Bearings.- Anti-friction bearings, in accordance with Specifications 42B5 and 42B10 as applicable, shall be used throughout, if practicable, and where not practicable, bronze bushings shall be used. Where bronze bushings are used positive means shall be provided to prevent the bushing from turning and cutting off the supply of lubricant. All bearings, except those lubricated by an oil bath shall be fitted with pressure type grease fittings in accordance with type I, III, IV or V of Specification MIL-F-3541.

3.2.9 Bedplates.- The bedplates shall be of adequate strength to maintain alignment without aid from the deck foundations. The holding down bolt spacing shall in general not exceed 15 times the bolt diameter and the thickness of the flange at these bolt holes shall be at least equal to the diameter of the bolts used. Foundation bolt holes shall be drilled 1/16 inch small for reaming out at installation to fit body bound bolts. The design shall eliminate pockets in which water can stand or collect.

3.2.10 Assembly.- The capstan shall be designed so that shafts, gears, bearings, and capstan head may be inspected, adjusted or removed without necessitating a complete dismantle or removal.

3.2.11 Motors.- Motors and motor characteristics shall be in accordance with Specification MIL-M-17059 or MIL-M-17556 as specified (see 6.1). One hundred percent of the motor load rating shall not be exceeded under the capstan rated load and speed. Unless otherwise specified in the contract or order, alternating current electric capstans shall be powered by two speed alternating current motors rated at full and one-half speeds.

3.2.12 Motor controllers.- Controllers and master switches shall be in accordance with Specification MIL-C-2174 or MIL-C-2212 as specified (see 6.1). An emergency run feature shall be provided in the control. Master switches shall be spring return to "OFF" and shall be deck mounted as near as practicable to the capstan they control. For direct current capstans, the control shall provide at least 3 speeds in both directions of rotation.

3.2.13 Electric brake.- Brakes shall be in accordance with Specification MIL-B-16392 as specified (see 6.1).

3.3 Performance.- The capstan shall be designed to perform as specified (see 6.1).

3.4 Identification plates.- Identification plates and other designated marking plates shall be in accordance with Specification MIL-I-15024.

3.5 Stress calculations.- A complete set of stress calculations, in duplicate, of gears, bearings, shafts, keys, holding down bolts, brakes, and power requirements shall be included with the first submission of drawings. The stress calculations shall contain a test or estimated speed-torque curve of the electric motor.

3.5.1 The stress calculations shall be presented in a neat clear form with references giving the source of formulas and information.

3.5.2 In stress calculations the following limitations shall exist:

3.5.2.1 Under rated conditions (see 6.1) the combined stresses in rotating parts shall not exceed 35 percent of the tensile yield point of the ductile materials used.

3.5.2.2 When the capstan head is subjected to the specified horizontal static load (see 6.1) acting at midheight of the capstan head, through and normal to the axis of the shaft, the combined stresses shall not exceed 70 percent of the tensile yield point in any part of the machine.

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3.5.2.3 When the capstan head is subjected to a pull equal to the stalling torque of the motor the combined stresses shall not exceed 70 percent in any part of the machine.

3.5.3 In stress calculations the following strength relationship shall exist:

Elastic strength	Percent of tensile yield point
Direct shear	60
Torsional shear	65
Compression (bearing)	160

3.5.3.1 Other valves may be used if substantiated by tests and approved by the bureau or agency concerned.

3.6 Submittal and approval of drawings.- All drawings shall conform to type II of Specification MIL-D-963, except that control drawings shall be type I. In addition, all electrical drawings shall conform to the applicable specification for the equipment. Prints for approval shall be submitted with an accompanying letter listing the drawings submitted by number and title and pointing out any departure from the requirements of this specification with the reasons therefor.

3.6.1 Drawings which are approved without exception may be released for manufacture at once. Drawings which are required to be modified shall be resubmitted for approval prior to manufacture unless the approving activity specifically grants permission to proceed with manufacture from altered drawings without resubmission.

3.6.2 Any work done prior to complete approval of the drawings shall be entirely at the risk of the contractor.

3.6.3 In addition to the assembly views of the unit, the general arrangement drawings shall contain a list of the mechanical onboard repair parts, weight of capstan, weight of repair parts, capstan ratings, and a complete list of reference drawings giving manufacturer's and bureau or agency drawing numbers and titles.

3.6.4 Two blueprint copies of all approved drawings shall be furnished to the cognizant Government inspector for use in conducting the inspection.

3.6.4.1 To expedite delivery the bureau or agency concerned may, at time of award, upon presentation of sufficient information to adequately describe construction, waive drawing approval prior to manufacture.

3.7 Instruction books.- Instruction books shall be in accordance with Specification MIL-B-15071. Type C, prints of the following drawings reduced for folding in a lengthwise direction only shall be made a part of the book:

(a) General arrangement drawing.
(b) Cross-sectional assembly drawings in sufficient number to adequately identify all parts.
(c) Bedplate or foundation drawing.
(d) Lubrication charts with lubricants listed according to applicable specification or symbol number.
(e) Operating instructions.
(f) Electric motor, controller and brake drawings.

A complete list of applicable drawings giving manufacturer's and bureau or agency drawing numbers and titles shall be included after the table of contents. The cover shall contain the applicable vessels listed after the title as on the title page. (See figure 2, Appendix 1, Specification MIL-B-15071.)

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3.8 Repair parts.-

3.8.1 Mechanical onboard.- Each set of mechanical onboard repair parts shall consist of the following items in sufficient quantity for one capstan unless otherwise specified in the contract or order:

(a) One set of each size and type of anti-friction bearing.
(b) One set of capstan head shaft bearing bushings.
(c) One set of intermediate shaft bearing bushings.
(d) One set of oil seals and special packings.

3.8.2 Electrical onboard.- Electrical onboard repair parts shall be furnished in accordance with the applicable equipment specification.

3.8.3 Special tools necessary for the proper maintenance of the equipment shall be included in each set of onboard repair parts. Each special tool shall be indelibly marked for the service for which it is intended.

3.8.4 Recommendations for onboard and stock repair parts shall be submitted as specified in Specification MIL-R-15137.

3.9 Finish.- Prior to shipment each capstan shall be cleaned and all parts not intended to remain bright shall be given two coats of primer conforming to Specification JAN-P-735 or JAN-P-72. A final coat of paint conforming to Specification MIL-P-15130 shall then be applied. Exposed surfaces that will be inaccessible after installation (underside of bedplates, etc.) shall be given three coats of primer conforming to Specification JAN-P-735 or JAN-P-72 or heavily coated with a rust preventive compound conforming to type A of Specification MIL-C-15167.

3.10 Workmanship.- The capstan and repair parts shall be free from all imperfections which might impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Tests.-

4.1.1 Tests at place of manufacture.- Capstans shall be subjected to the following tests at the place of manufacture:

4.1.1.1 Each unit shall be run under no load for a period of one-half hour in each direction and observed to determine that no abnormal heating wear or noise shall occur.

4.1.1.2 One unit shall be selected from each lot offered for delivery and tested at rated load at rated speed for one-half hour in both directions of rotation. It shall also be tested to the specified strength of the rope at no specified speed without stalling the motor. The electric brake shall be tested by stopping and holding the capstan with 150 percent rated line pull.

4.1.1.3 The tests specified in 4.1.1.1 and 4.1.1.2 shall be performed under the observation of the Government inspector and test data recorded as necessary to demonstrate satisfactory performance.

4.1.2 Shipboard tests.- After installation aboard ship each capstan will be tested as specified in 4.1.1.1 and 4.1.1.2 to determine conformance with specified requirements.

4.2 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. -

5.1.1 For domestic shipment - immediate use. - Capstans and accessories shall be preserved and packaged in accordance with the manufacturer's commercial practice and packed in a manner to insure acceptance and safe delivery at destination. Containers shall comply with the Consolidated Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.1.2 For domestic shipment and storage. - Each capstan shall be preserved and packaged in accordance with Specification MIL-P-3184 and packed in unsheathed crates conforming to Specification MIL-C-3774 or JAN-P-132 as applicable. Capstans shall be shrouded within unsheathed crates with waterproof barrier material conforming to Specification JAN-P-125. Detached components shall be packed in domestic type cleated plywood or nailed wood boxes conforming to Specification PPP-B-601 or NN-B-621 and, when practicable, shall be secured within the crate holding the capstan.

5.1.3 For overseas shipment. - Each capstan shall be preserved and packaged in accordance with Specification MIL-P-3184 and packed in sheathed crates conforming to Specification JAN-P-104 or MIL-C-3769 as applicable. Detached components shall be packed in overseas type cleated plywood or nailed wood boxes conforming to Specification PPP-B-601 or JAN-P-106 and, when practicable, shall be packed within the crate containing the winch. Boxes shall be lined with a sealed waterproof caseliner conforming to Specification MIL-L-10547 and shall be closed and strapped in accordance with the appendix of the applicable container specification.

5.1.3.1 When specified (see 6.1), capstans shall be packed in unsheathed crates conforming to Specification MIL-C-3774 or JAN-P-132 as applicable, and shall be shrouded within crates with waterproof barrier material conforming to Specification JAN-P-125.

5.1.4 Repair parts and publications. - Onboard and stock repair parts and publications shall be preserved, packaged and packed for domestic shipment and storage or overseas shipment, as specified (see 6.1), in accordance with Specification MIL-P-17652 or MIL-P-16298, as applicable, or in accordance with Specification MIL-P-116 for items not covered therein when specified in the contract or order. Onboard repair parts shall be packaged in intermediate repair parts boxes conforming to Specification MIL-B-233 and overpacked as specified in Specification MIL-P-17652.

5.2 Marking. - Marking of the packages and shipping containers shall be in accordance with Standard MIL-STD-129. Nomenclature shall be the exact nomenclature approved under the contract or order.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Arrangement of components (see 3.2.1). For 3.2.1 (a) arrangement give distance between bedplate mounting surfaces.
(c) Rope size (see 3.2.6).
(d) Characteristics of motor (see 3.2.11).
(e) Characteristics of controller (see 3.2.12).
(f) Characteristics of brake (see 3.2.13).

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(g) Performance requirements (see 3.3):
  (1) Rated load and line speed (see 3.5.2.1).
  (2) Maximum load at no specified speed (see 3.5.2.1).
  (3) Static load at midheight of capstan head (see 3.5.2.2).
(h) Whether the equipment will be prepared for domestic shipment - immediate use; domestic shipment and storage; or overseas shipment (see 5.1).
(i) Each bid should include the following data, if applicable. Failure to comply with this requirement may result in disqualification of bid:
  (1) An approximate dimensioned general arrangement drawing and diagrammatic drawing showing gearing and component parts of the capstan.
  (2) Estimated weight of capstan and electrical components.
  (3) Manufacturer's data for proposed motor, controller and brake.

Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Bureau of Ships
Other interest:
None

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commander
Naval Ship Engineering Center
Center Building
Prince George's Center
Hyattsville, Maryland 20782

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PLATE NO. 15419 (BACK)

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